

TRINITY INTERNATIONAL COLLEGE

Dillibazar Height, Kathmandu

Final Assessment 2025

BCA (Batch-2080)

SET B

Semester: Third

Subject: Data Structure & Algorithm

Course No: CSCS 201

Full Marks: 50

Pass Marks: 24

Time: 2hrs 50min

Candidates are required to give answers in their own words as far as practicable. The figures in the margin indicate full marks.

Subjective

Group 'B'

Attempts Any Six Questions

[6×5=30]

1. Define data structure. How is a stack different from a queue? Explain.
2. List any two applications of a stack. Convert the infix expression $A+B-(C*D+A)/(C-D)+E$ into postfix notation using a stack.
3. Define a priority queue. How is a circular queue different from a double-ended queue? Explain.
4. What is a deterministic algorithm? Differentiate between serial and parallel algorithms.
5. What is collision in hashing? How can collision resolution be done using the separate chaining method? Explain.
6. How does quicksort work? Write a function to perform quicksort.
7. Create a Huffman Tree and compute Huffman codes for the following characters along with their frequencies using the Huffman algorithm:

Characters	A	E	I	O	U	S	Q
Frequencies	10	15	12	3	4	13	1

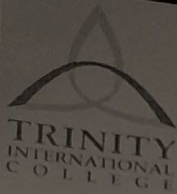
Group 'C'

Attempt Any Two Questions.

[2×10=20]

8. Explain different types of graphs. Explain Kruskal's algorithm in detail.
9. Explain balanced binary trees and AVL trees with proper examples. Create an AVL tree from the following sequence:
4, 6, 12, 19, 15, 25, 13, 8, 3, 7, 11, 17, 31.
10. How is binary search different from sequential search? Explain. Consider a hash table of size 9 and insert the keys 9, 19, 32, 10, 73, 85, 50 using linear probing.

"Good Luck"



TRINITY INTERNATIONAL COLLEGE

Dillibazar Height, Kathmandu

Final Assessment 2025

BCA (Batch-2080)

SET A

Semester: Third

Subject: Probability & Statistics

Course No: CSCS 202

Full Marks: 50

Pass Marks: 24

Time: 2hrs 50min

Candidates are required to give answers in their own words as far as practicable. The figures in the margin indicate full marks.

Group 'B'

Attempts any six Questions

[6×5=30]

1. Determine average wages and standard deviation of wages from the following data:

Wages	25 - 30	30 - 35	35 - 40	40 - 45	45 - 50	50 - 55
No. of Workers	10	13	18	21	24	28

2. The following data represents the marks of 100 students. If the mode value is 58, find the missing frequencies.

Marks	0-20	20-40	40-60	60-80	80-100
No. of students	14	?	27	?	15

3. The mean of the marks obtained in the exam by a group of 90 students was found to be 49.96. The mean of the marks obtained by the same exam by another group of 150 students was 52.32. Find the mean of the marks obtained by both groups of students taken together.

4. A manager of a large company wants to check the inventory records against the physical inventories through a sample survey. He wants to ensure that the maximum sampling error should not be more than 5% above or below the true population of the inaccurate records. The proportion of inaccurate records is estimated at 25% from past experiences.

- Determine sample size.
- How large a sample should be taken when such estimates from experience are not available?

5. The following table contains information on the amount of time spent each day (on average) on social networks and the Internet for social or entertainment purposes and the grade point average for a random sample of 9 college students. Here GPA is the dependent variable

Time (hours per day)	1.4	5.0	5.2	3.4	2.5	6.1	1.3	4.2	3.9
GPA	3.91	3.20	3.15	3.3	3.7	3.0	3.95	3.4	3.5

- Find the simple correlation coefficient
 - Is the correlation significant?
 - Obtain the range of the population correlation coefficient.
- Write short notes on errors in statistics.
 - Write down the importance of statistics in computer applications.

Attempt Any Two Questions.

Group 'C'

[2x10=20]

8. The length of life (in hours) of two brands of laptop batterybackup is given below.

HP	5.5	4.8	5.3	3.6	5.0	5.8	6.1	5.0	6.0
Dell	4.5	4.3	3.0	5.1	6.5	4.0	4.5	6.0	6.5

Show which brand's laptop batterybackup is more consistent based on the coefficient of variation.

9. General manager of the New Laptop Shop of Baneshwor, is concerned about the sales behavior of Lenovo brand laptops sold at the shop. He realizes that many factors might help the sales, among price major ones. Following data recorded.

y (Sales unit)	50	40	70	80	60	55
x (Price in '000 Rs)	70	50	80	90	100	45

- Construct a scatter plot.
 - Estimate the simple regression equation that best fits the data.
 - Predict the value of sales when the price is Rs: 75000.
- 10 The response time in milliseconds was determined for three different types of circuits in an electronic calculator. The results are shown in the following table.

Circuit Type	Response Time				
	19	22	20	18	25
A	19	22	20	18	25
B	20	21	33	27	40
C	16	15	18	26	17

Using $\alpha = 0.01$, test the hypothesis that the three circuit types have the same response time.

"Good Luck"



Dillibazar Height, Kathmandu

Final Assessment 2025

BCA (Batch - 2080)

SET A

Semester: Third

Subject: Web Technology

Course No: CACS 205

Full Marks: 50

Pass Marks: 24

Time: 2.50hrs

Candidates are required to give their answers in their own words as far as practicable. The figure in the margin indicates full marks.

Group B

Attempt any six questions.

[6x5 = 30]

1. Explain the importance of HTML tables in web design. Describe how to align content within a table cell with an example.
2. Create an HTML form to collect user information, including name, email, gender (using radio buttons), and favorite programming language (using a dropdown). Explain the purpose of each form element used in your code.
3. What is a cookie? Describe cookie with example.
4. What is CSS Selector? Explain different types of selectors with suitable example.
5. Why session is required in web development? Explain how you set and remove values stored in session with a suitable example.
6. Explain the concept of tier technology in web applications. Compare 2-tier and n-tier architectures with examples.
7. What is error handling? What are different technique of error handling with example?
8. Write a short note on any two of the following:
 - a. XML Namespaces
 - b. Dynamic Content Generation
 - c. Lists

Group C

[2x10 = 20]

- Attempt any two questions.
1. Explain how to connect a PHP web application to a database. Write a sample PHP code to establish a connection and retrieve data from a table.
 2. What are the key components of an XML file? Write an XML document to represent a "book" element with attributes such as title, author, and ISBN, and define an XSD schema and DTD to validate it.
 3. Create a registration form using HTML and style it using CSS. Your form should include the following fields:
 - Full Name ✓
 - Email Address ✓
 - Password ✓
 - Gender (radio buttons) ✓
 - Hobbies (checkboxes) ✓
 - A Submit button
 - A Reset button
 - Use proper label and placeholder.
 - Write a SQL query for these fields.

align
text-align

valign
vertical-align

Semester: Third

Full Marks: 50

Subject: Object Oriented Programming

Pass Marks: 24

Course No: CACS 204

Time: 2:50hr

Candidates are required to give their answers in their own words as far as practicable. The figure in the margin indicates full marks.

Subjective Test

Group B

Attempt ANY SIX Questions:

[6x5=30]

1. How does the Java Virtual Machine (JVM) contribute to platform independence?
Write a Java code to print "Hello, JVM". [2 + 3]
2. Differentiate between while, do-while, and for loops in Java. Provide code for each to print numbers 1 to 5. [2 + 3]
3. Write a program to demonstrate the use of super keyword and method overriding in Java. [2 + 3]
4. What is the significance of finally block? Write a program to demonstrate try-catch-finally for dividing two numbers. [2 + 3]
5. What is the use of StringBuffer class? Write code to reverse a string using for StringBuffer. [2 + 3]
6. Write a code that reads a file line by line and prints its contents to the console using BufferedReader. [2 + 3]
7. Write short notes on (Any Two): a) Serialization and Deserialization
b) JTable in Swing
c) Autoboxing and Unboxing
[2.5 x 2 = 5]

SP SUM C
String v = "Hello";
String rev = "olleH";
for (i = v.length() - 1; i >= 0; i--)
rev = charAt(i) + rev;
[2x10=20]
Sout rev

Group C

Attempt ANY TWO Questions:

1. Create a simple banking system with classes Account, SavingsAccount, and CurrentAccount. Demonstrate inheritance, method overriding, and polymorphism.
[10 marks]
2. Write a Java application using JDBC to connect to a database College, insert a student record into the Students table (columns: ID, Name, Department, CGPA), and retrieve all students with CGPA > 3.5. [2 (JDBC intro) + 8 (code)]
3. Design a GUI-based calculator using Swing that performs basic arithmetic operations (add, subtract, multiply, divide) using buttons and displays the result in text field. [10 marks]

while do while
String Buffer
Super method overriding



Final Assessment 2025

BCA (Batch-2080)

Semester: Third

Subject: System Analysis and Design

Course No: CACS 203

SET - A

Full Marks: 50

Pass Marks: 24

Time: 3 Hrs.

Candidates are required to give answers in their own words as far as practicable. The figures in the margin indicate full marks.

Group 'B'

Attempt any SIX questions.

[6×5=30]

2. Define information systems analysis and design. Explain its importance in organizational improvement. (2+3)
3. List and briefly explain the phases of the Systems Development Life Cycle (SDLC). *Planning, Analysis, Design, Implementation, Maintenance*
4. Differentiate between logical design and physical design in the SDLC. (2+3)
5. What are CASE tools? Explain any two types of CASE tools with examples. (2+3)
6. Compare Agile Methodologies with traditional Waterfall SDLC in terms of user involvement.
7. Explain the concept of corrective maintenance with an example. How does it differ from adaptive maintenance? (3+2)
8. What is the role of a system librarian in configuration management?
9. Describe the importance of quality documentation in reducing maintenance efforts.

Group C

Attempt any TWO questions.

[2×10=20]

10. Explain the key principles of Agile Methodologies as outlined in the Agile Manifesto. Why are these methodologies suitable for projects with dynamic requirements? (6+4)
11. Draw a context diagram and a level-1 Data Flow Diagram (DFD) for a library management system that handles book borrowing and returning. (5+5)
12. Discuss the process of maintaining information systems, including the four major activities involved. How does configuration management ensure only authorized changes are made? (6+4)

GOOD LUCK

*Trinity
Research program
High productivity
Flexibility
Reduced development time*

*Clearly defined stage
Documentation
Early Test organization
System Analyst*

SAD/IC/20MAY2025

*While developing
any better
or method avoid*